Gargantua's disk looks quite different from the pictures of thin disks that you see on the web or in astrophysicists' technical publications, because those pictures omit a key feature: the gravitational lensing of the disk by its black hole. Not so in *Interstellar*, where Chris insisted on visual accuracy.

Eugénie von Tunzelmann was charged with putting an accretion disk into Oliver James' gravitational lensing computer code, the code I described in Chapter 8. As a first step, just to see what the lensing does, Eugénie inserted a disk that was truly infinitesimally thin and lay precisely in Gargantua's equatorial plane. For this book she has provided a more pedagogical version of that disk, made of equally spaced color swatches (Inset in Figure 9.7).

If there had been no gravitational lensing, the disk would have looked like the inset. The lensing produced huge changes from this (body of Figure 9.7). You might have expected the back portion of the disk to be hidden behind the black hole. Not so. Instead, it is gravitationally lensed to produce two images, one above Gargantua and the other below; see Figure 9.8. Light rays emitted from the disk's top face, behind Gargantua, travel up and over the hole to the camera, producing the disk image that wraps over the top of Gargantua's shadow in Figure 9.7; and similarly for the disk image that wraps under the bottom of Gargantua's shadow.